

Learning Objectives

Advanced Literature Search

- Advanced Literature Searching
 - Known which are the best tools to search for literature
 - Understand the logic of constructing advanced search queries
 - Know which basic search tags to use for generating specific search queries
 - Conduct more efficient searches using PubMed

Learning Outcomes

- Advanced Literature Searching
 - Locate and understand the advanced search pages for PubMed
 - Generate complex search queries using provided examples
 - Use some of the major search fields (tags) to generate specific search results
 - Further refine your search results for specific articles

Literature Searching



- Essential aspect of all research
- Most resources are now electronically available
- Not necessarily easier to retrieve
- Searching the literature can be a challenging task

Google

Google can bring you back
100,000 answers
A LIBRARIAN can bring you back the
RIGHT ONE
– Neil Gaiman

Useless Results

Google What am I searching for

All Images Videos News Maps More Search tools

Page 10 of about 67 400 000 results (0,54 seconds)

How to Search on Google: 31 Advanced Google Search Tips - HubSpot
blog.hubspot.com/blog/.../12-Quick-Tips-To-Search-Google-Like-An-Expert.aspx
Nov 24, 2015 - Uncover 31 advanced search tips for conducting better Google ... until you find what you're looking for, then I'm here to tell you that there's a ...

District/Official Search - Illinois State Board of Elections
www.elections.state.il.us/districtlocator/districtofficialsearchbyaddress.aspx
Instructions: Select either desktop or mobile/Lite search applications below to search for district/officials. If you are having problems loading the Silverlight ...

Easy Search
easysearch.lib.cuhk.edu.hk/
Easy Search is a one-stop search for discovery and delivery of local and ... What am I searching?
Search. All. Search for all types of materials listed below.

Answer 6 Questions to Reveal Your Life Purpose | SUCCESS
www.success.com/article/answer-6-questions-to-reveal-your-life-purpose
Dec 25, 2014 - Get ready for some soul-searching. ... I, the President of the U.S., (your name) am designed to be a good communicator (insert personal quality) ...

Useless Results

How do i delete the trending items on my google search bar history ...
productforums.google.com/d/topic/websearch/FtSE0mP1qyo
Jan 29, 2016 - 100+ posts - 100+ authors
Andy Wiles, 1/30/16 11:54 AM. I really hope somebody finds a way to fix this issue! I keep my search history cleared and nice and tidy, except ...

Patent Searching 101: A Patent Search Tutorial - IPWatchdog.com ...
www.ipwatchdog.com/2015/07/11/patent-searching-101-a...search...2?id=59308/
Jul 11, 2015 - It is quite common for inventors to do a patent search and find nothing even when ...
Gene, I am pleased to see that you keep emphasizing the ...

Searching for Eternity - Page 8 - Google Books Result
<https://books.google.co.za/books?isbn=076420372X>
Elizabeth Musser - 2007 - Fiction
Of course I am thinking about him, but at the same time, I am thinking of her, wondering if she is down in Bolivia, if she helped them locate the Butcher. "You may ...

Searches related to What am I searching for

searching for **something quotes** what am i searching for **quiz**
what am i searching for **in life** i **don't know** what i'm looking for **lyrics**
what am i searching for **on google** i'm searching for **something lyrics**
searching for **something more in life** i **have everything but still something is missing quotes**

< **Gooooooooooooo**gle >
Previous 5 6 7 8 9 10 11 12 13 14 Next

this lecture should save you from th



Popular Literature Resources

- Google Scholar – Specialised google search engine for academic literature
- PubMed – bibliographical database run by the NCBI covering a range of biomedical literature including MEDLINE
- PubMed Clinical Queries – A specialised PubMed search for searching for clinical data

MEDLINE



- The US National Library of Medicine (NLM) is the world's largest medical library
- NLM has been indexing biomedical literature since 1897 to help provide easy access to literature
- The printed index to articles, *Index Medicus*, was converted into the database known today as MEDLINE

Other literature resources

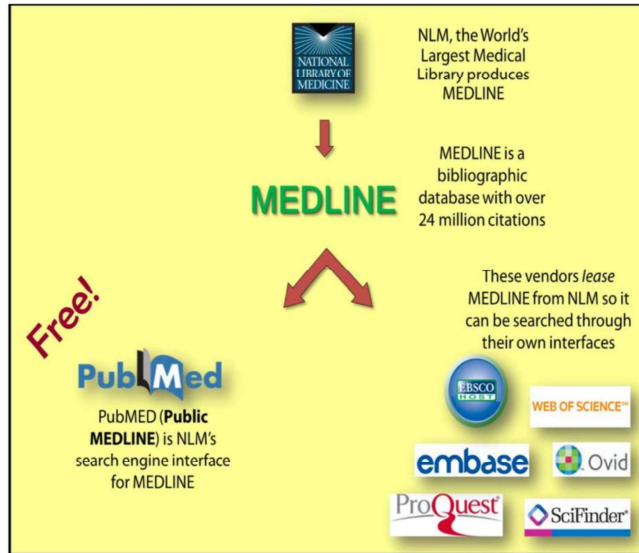
- Scopus – largest abstract and citation database of peer-reviewed literature across all scientific domains
- ScienceDirect – Linked to Scopus and Elsevier, search across a number of journals from all scientific domains

MEDLINE



- Since 1996, free access to MEDLINE has been made available to the public via PubMed at NCBI
- In addition to MEDLINE PubMed contains citations from additional biomedical journals and books.
- PubMed provides access to more than 26 million citations for biomedical literature

MEDLINE



<http://research.library.gsu.edu/pubmed>



Why use PubMed/MEDLINE

- Comprehensive, current up-to-date information
- Literature from credible sources
- Controlled vocabulary for indexing and searching
- Abstract and/or full text often provided
- Can filter search results to narrow query

Literature Searching Steps

- First step is to clearly know what information you are looking for
- Break down your search question into keywords
- Identify the main sources for quality data
- Decide if it is best to combine your keywords or search for them separately

<http://www.ncbi.nlm.nih.gov/pubmed/>

ADVANCED SEARCHING USING PUBMED

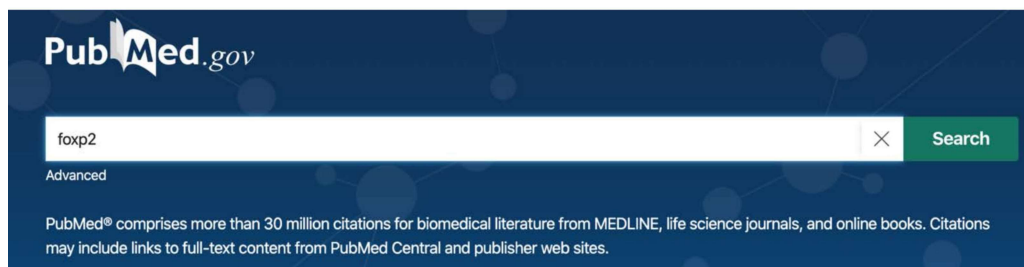
Example: FOXP2

- You are interested in studying genetic variation in *FOXP2* and its impact of speech development
- You would like to review all studies that have linked mutations in *FOXP2* to some sort of speech development disorder
- You would also like to find the original case study that suggested that *FOXP2* is responsible for speech development in humans

Breaking down the search query

- You are interested in studying **genetic variation** in ***FOXP2*** and its impact of **speech development**
- You would like to review all studies that have linked **mutations** in ***FOXP2*** to a **speech development disorder**
- You would also like to find the original study by Lai *et. al.* in 2001 that suggested that ***FOXP2*** is responsible for **speech development** in **humans**

Simplest Search



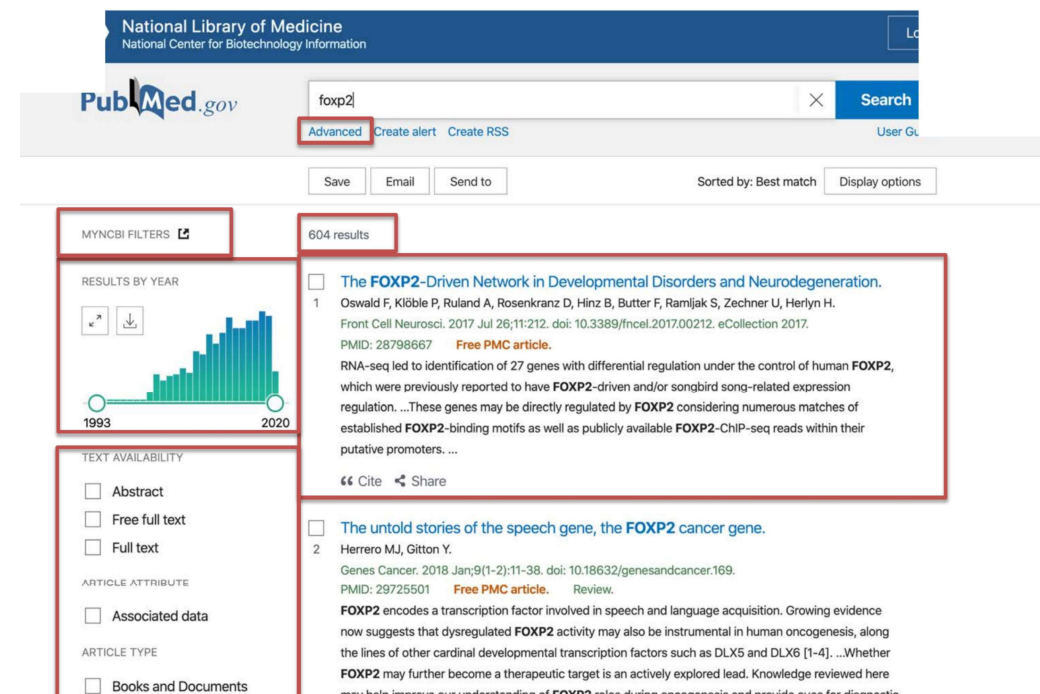
PubMed.gov

foxp2

Advanced

Search

PubMed® comprises more than 30 million citations for biomedical literature from MEDLINE, life science journals, and online books. Citations may include links to full-text content from PubMed Central and publisher web sites.



National Library of Medicine
National Center for Biotechnology Information

PubMed.gov

foxp2

Advanced Create alert Create RSS

Save Email Send to Sorted by: Best match Display options

MYNCBI FILTERS

RESULTS BY YEAR

1993 2020

TEXT AVAILABILITY

☐ Abstract

☐ Free full text

☐ Full text

ARTICLE ATTRIBUTE

☐ Associated data

ARTICLE TYPE

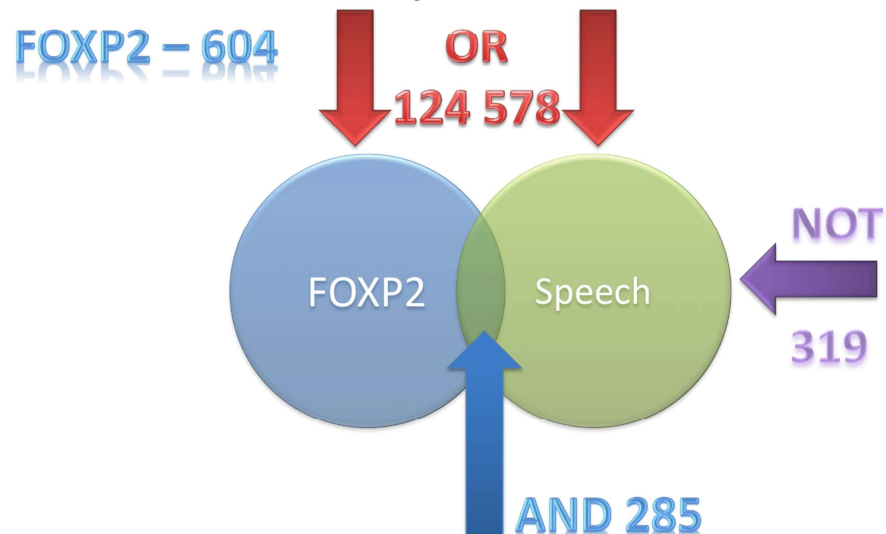
☐ Books and Documents

604 results

☐ 1 The **FOXP2**-Driven Network in Developmental Disorders and Neurodegeneration.
Oswald F, Klöble P, Ruland A, Rosenkranz D, Hinz B, Butter F, Ramjak S, Zechner U, Herlyn H.
Front Cell Neurosci. 2017 Jul 26;11:212. doi: 10.3389/fncel.2017.00212. eCollection 2017.
PMID: 28798667 [Free PMC article.](#)
RNA-seq led to identification of 27 genes with differential regulation under the control of human **FOXP2**, which were previously reported to have **FOXP2**-driven and/or songbird song-related expression regulation. ...These genes may be directly regulated by **FOXP2** considering numerous matches of established **FOXP2**-binding motifs as well as publicly available **FOXP2**-ChIP-seq reads within their putative promoters. ...
Cite Share

☐ 2 The untold stories of the speech gene, the **FOXP2** cancer gene.
Herrero MJ, Gitten Y.
Genes Cancer. 2018 Jan;9(1-2):11-38. doi: 10.18632/genesandcancer.169.
PMID: 29725501 [Free PMC article.](#) Review.
FOXP2 encodes a transcription factor involved in speech and language acquisition. Growing evidence now suggests that dysregulated **FOXP2** activity may also be instrumental in human oncogenesis, along the lines of other cardinal developmental transcription factors such as DLX5 and DLX6 [1-4]. ...Whether **FOXP2** may further become a therapeutic target is an actively explored lead. Knowledge reviewed here may help improve our understanding of **FOXP2** roles during oncogenesis and provide cues for diagnostic.

Combining Search Terms – Boolean Operators



Other search operators

- Truncation
 - Child* = Child and children
- Quotations
 - “copy number variation”

Advanced Search Builder

National Library of Medicine
National Center for Biotechnology Information

PubMed Advanced Search Builder

Add terms to the query box

All Fields

Query box

History and Search Details

Your history is currently empty! As you use PubMed, your history will build up here.

Search

FIELD NAME	[TAG]	SEARCHES
Affiliation	[AD]	First listed author's institutional affiliation and address
Author	[AU]	Author name
Corporate Name as Author	[CN]	Corporate names as authors
Full Author Name	[FAU]	Full author name (2002 onward, if available)
Issue	[IP]	Journal issue number
Journal Title	[TA]	Title of journal, journal title abbreviation, ISSN
Language	[LA]	Language of the article (not the abstract)
Last Author Name	[LASTAU]	Last personal author in a citation
MeSH Major Topic	[MAJR]	Main topic of an article
MeSH Subheading	[SH]	MeSH Subheading
MeSH Terms	[MH]	MeSH Term
Pagination	[PG]	First page number of a journal article
Personal Name as Subject	[PS]	Person as the subject of an article, not as an author
Publication Date	[PD]	Date an article was published
Publication Type	[PT]	Format of an article (letter, clinical trial, etc) rather than content
Substance Name	[NM]	Chemical and substance names discussed in an article
Text Word	[TW]	Textual fields of PubMed records
Title	[TI]	Article title
Title or Abstract	[TIAB]	Words in an article title or an abstract
Volume	[VI]	Volume number of a particular journal
Year	[DP]	The year when a journal article was published. Search a date range, type in 2000:2009[DP]

Advanced Search Query

NIH National Library of Medicine
National Center for Biotechnology Information

Log in

PubMed Advanced Search Builder

PubMed.gov
User Guide

Add terms to the query box

All Fields Enter a search term ADD Show Index

Query box

Enter / edit your search query here Search

History and Search Details

Your history is currently empty! As you use PubMed your recent searches will appear here.

Breaking down the search query

- You are interested in studying genetic variation in *FOXP2* and its impact of speech development
- You would like to review all studies that have linked mutations in *FOXP2* to a speech development disorder
- You would also like to find the original study by Lai *et. al.* in 2001 that suggested that *FOXP2* is responsible for speech development in humans

FOXP2 Query Building

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User Guide

Add terms to the query box

All Fields Enter a search term ADD Show Index

Query box

Enter / edit your search query here Search

History and Search Details

Download Delete

Search	Actions	Details	Query	Results	Time
#3	...	>	Search: mutation	1,093,123	04:58:47
#2	...	>	Search: speech development	30,109	04:58:34
#1	...	>	Search: foxp2	604	04:58:05

Showing 1 to 3 of 3 entries

FOXP2 Query Building

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National Center for Biotechnology Information

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PubMed Advanced Search Builder

PubMed.gov
User Guide

Add terms to the query box

All Fields Enter a search term AND Show Index

Query box

((foxp2) AND (speech development)) AND (mutation) Search

History and Search Details

Download Delete

Search	Actions	Details	Query	Results	Time
#5	...	>	Search: ((foxp2) AND (speech development)) AND (mutation)	92	05:03:54
#3	...	>	Search: mutation	1,093,123	04:58:47
#2	...	>	Search: speech development	30,109	04:58:34
#1	...	>	Search: foxp2	604	04:58:05

Showing 1 to 4 of 4 entries

PubMed.gov

((foxp2) AND (speech development)) AND (mutation)

Advanced Create alert Create RSS User Guide

Save Email Send to Sorted by: Best match Display options

MYNCBI FILTERS 12 92 results

RESULTS BY YEAR

TEXT AVAILABILITY

☐ Abstract

☐ Free full text

☐ Full text

ARTICLE ATTRIBUTE

☐ Associated data

ARTICLE TYPE

☐ Books and Documents

☐ Clinical Trial

☐ Meta-Analysis

☐ Randomized Controlled Trial

☐ Review

☐ Systematic Reviews

1 **The Key Regulator for Language and Speech Development, FOXP2, is a Novel Substrate for SUMOylation.**

Meredith LJ, Wang CM, Nascimento L, Liu R, Wang L, Yang WH. J Cell Biochem. 2016 Feb;117(2):426-38. doi: 10.1002/jcb.25288. PMID: 26212494 [Free PMC article.](#)

Transcription factor forkhead box protein P2 (FOXP2) plays an essential role in the development of language and speech. However, the transcriptional activity of FOXP2 regulated by the post-translational modifications remains unknown. ...Interestingly, we observed that human etiological FOXP2 R553H mutation robustly reduces its SUMOylation potential as compared to wild-type FOXP2. ...

“ Cite Share

2 **FOXP2 variants in 14 individuals with developmental speech and language disorders broaden the mutational and clinical spectrum.**

Reuter MS, Riess A, Moog U, Briggs TA, Chandler KE, Rauch A, Stampfer M, Steindl K, Gläser D, Joset P; DDD Study, Krumbiegel M, Rabe H, Schulte-Mattler U, Bauer P, Beck-Wödl S, Kohlhasse J, Reis A, Zweier C. J Med Genet. 2017 Jan;54(1):64-72. doi: 10.1136/jmedgenet-2016-104094. Epub 2016 Aug 29. PMID: 27572252

RESULTS: We identified four different truncating mutations, two novel missense mutations within the forkhead domain and an intragenic deletion in FOXP2 in 14 individuals from eight unrelated families. ...Motor development was normal or only mildly delayed. Mild cognitive impairment was reported for most individuals. CONCLUSIONS: By identifying intragenic deletions or mutations in 14 individuals from eight unrelated families with variable developmental delay/cognitive impairment and speech and language deficits, we considerably broaden the mutational and clinical spectrum associated with aberrations in FOXP2. ...

“ Cite Share

MYNCBI FILTERS 12 3 results

RESULTS BY YEAR

TEXT AVAILABILITY

☐ Abstract

☐ Free full text

☐ Full text

ARTICLE ATTRIBUTE

☐ Associated data

ARTICLE TYPE

☐ Books and Documents

☐ Clinical Trial

☐ Meta-Analysis

☐ Randomized Controlled Trial

☐ Review

☐ Systematic Reviews

PUBLICATION DATE

☐ 1 year

☐ 5 years

☐ 10 years

Additional filters

1 **The Key Regulator for Language and Speech Development, FOXP2, is a Novel Substrate for SUMOylation.**

Meredith LJ, Wang CM, Nascimento L, Liu R, Wang L, Yang WH. J Cell Biochem. 2016 Feb;117(2):426-38. doi: 10.1002/jcb.25288. PMID: 26212494 [Free PMC article.](#)

Here, we demonstrated that FOXP2 is clearly defined as a SUMO target protein at the cellular levels as FOXP2 is covalently modified by both SUMO1 and SUMO3. ...Interestingly, we observed that human etiological FOXP2 R553H mutation robustly reduces its SUMOylation potential as compared to wild-type FOXP2. ...

“ Cite Share

2 **An aetiological Foxp2 mutation causes aberrant striatal activity and alters plasticity during skill learning.**

French CA, Jin X, Campbell TG, Gerfen E, Groszer M, Fisher SE, Costa RM. Mol Psychiatry. 2012 Nov;17(11):1077-85. doi: 10.1038/mp.2011.105. Epub 2011 Aug 30. PMID: 21876543 [Free PMC article.](#)

Mutations in the human FOXP2 gene cause impaired speech development and linguistic deficits, which have been best characterised in a large pedigree called the KE family. ...Independent studies in multiple species suggest that the striatum is a key site of FOXP2 action. Here, we used in vivo recordings in awake-behaving mice to investigate the effects of the KE-family mutation on the function of striatal circuits during motor-skill learning. ...

“ Cite Share

3 **Foxp2 mutations impair auditory-motor association learning.**

Kurt S, Fisher SE, Ehret G. PLoS One. 2012;7(3):e33130. doi: 10.1371/journal.pone.0033130. Epub 2012 Mar 7. PMID: 22412993 [Free PMC article.](#)

The magnitude of impairments in association learning, however, depends on the nature of the mutation. Mice with a missense mutation in the DNA-binding domain are able to learn, but at a much slower rate than wild type animals, while mice carrying an early nonsense mutation learn very little. ...Given the importance of such networks for the acquisition of human spoken language, and the fact that similar mutations in human FOXP2 cause problems with speech development, this work opens up a new perspective on the use of mouse models for understanding pathways underlying speech and language disorders. ...

Narrower search

MeSH Terms

Other Term

Pagination

Pharmacological Action

Publication Type

Publisher

Secondary Source ID

Subject - Personal Name

Supplementary Concept

Text Word

Title

✓ Title/Abstract

Transliterated Title

Volume

Query box

foxp2[Title/Abstract]

Search

History and Search Details

Download Delete

Search	Actions	Details	Query	Results	Time
#4	...	>	Search: ((foxp2[Title/Abstract]) AND (speech development[Title/Abstract])) AND (mutation[Title/Abstract])	3	05:10:29
#3	...	>	Search: mutation[Title/Abstract]	358,784	05:10:01
#2	...	>	Search: speech development[Title/Abstract]	1,079	05:09:44
#1	...	>	Search: foxp2[Title/Abstract]	555	05:09:27

Showing 1 to 4 of 4 entries

MeSH Terms

- Medical Subject Headings
 - Set of controlled vocabulary terms used by the NLM
 - Articles are assigned MeSH terms by subject specialists who read over the articles and determine the subject contents
 - Ensures that articles on the same subject are grouped together regardless of the terms used by the authors

MeSH Terms

- May be useful to look up the MeSH term for your keywords in order to run a more specific search

MeSH Terms

MeSH [Create alert](#) [Limits](#) [Advanced](#)

Summary ▼ Send to: ▼

Search results

Items: 2

☐ [Language Development Disorders](#)
1. Conditions characterized by language abilities (comprehension and expression of **speech** and writing) that are below the expected level for a given age, generally in the absence of an intellectual impairment. These conditions may be associated with DEAFNESS; BRAIN DISEASES; MENTAL DISORDERS; or environmental factors.
Year introduced: 1986(1980)

☐ [Speech Development, Delayed, With Facial Asymmetry, Strabismus, And Transverse Earlobe Crease \[Supplementary Concept\]](#)
2. Date introduced: November 5, 2012

MeSH Terms

MeSH [Create alert](#) [Limits](#) [Advanced](#)

[All MeSH Categories](#)

[Diseases Category](#)

[Nervous System Diseases](#)

[Neurologic Manifestations](#)

[Neurobehavioral Manifestations](#)

[Communication Disorders](#)

[Language Disorders](#)

Language Development Disorders

[Specific Language Disorder](#)

FOXP2 + MeSH

PubMed Advanced Search Builder PubMed.gov [User Guide](#)

Add terms to the query box

[Show Index](#)

Query box

History and Search Details Download Delete

Search	Actions	Details	Query	Results	Time
#6	...	>	Search: ((foxp2[Title/Abstract]) AND (language development disorders[Title/Abstract])) AND (mutation[Title/Abstract])	13	05:32:40
#4	...	>	Search: ((foxp2[Title/Abstract]) AND (speech development[Title/Abstract])) AND (mutation[Title/Abstract])	3	05:23:41
#3	...	>	Search: mutation[Title/Abstract]	358,784	05:10:01
#2	...	>	Search: speech development[Title/Abstract]	1,079	05:09:44
#1	...	>	Search: foxp2[Title/Abstract]	555	05:09:27

Showing 1 to 5 of 5 entries

Your search was processed without automatic term mapping because it retrieved zero results.

- ☐ **Equivalent missense variant in the *FOXP2* and *FOXP1* transcription factors causes distinct neurodevelopmental disorders.**
- 1 Sollis E, Deriziotis P, Saltsu H, Miyake N, Matsumoto N, Hoffer MJV, Ruivenkamp CAL, Alders M, Okamoto N, Bijlsma EK, Plomp AS, Fisher SE.
Hum Mutat. 2017 Nov;38(11):1542-1554. doi: 10.1002/humu.23303. Epub 2017 Aug 14.
PMID: 28741767
- The closely related paralogues **FOXP2** and **FOXP1** encode transcription factors with shared functions in the **development** of many tissues, including the brain. ...We present here for the first time a direct comparison of the molecular and clinical consequences of the same **mutation** affecting the equivalent residue in **FOXP1** and **FOXP2**. ...
- 🗨 Cite ➦ Share
- ☐ **Early neuroimaging markers of *FOXP2* intragenic deletion.**
- 2 Liégeois FJ, Hildebrand MS, Bonthron A, Turner SJ, Scheffer IE, Bahlo M, Connelly A, Morgan AT.
Sci Rep. 2016 Oct 13;6:35192. doi: 10.1038/srep35192.
PMID: 27734906 **Free PMC article.**
- FOXP2** is the major gene associated with severe, persistent, developmental speech and **language disorders**. While studies in the original family in which a **FOXP2** **mutation** was found showed volume reduction and reduced activation in core **language** and speech networks, there have been no imaging studies of different **FOXP2** mutations. ...The hippocampus is implicated for the first time in **FOXP2** diseases. We conclude that **FOXP2** anomaly is either directly or indirectly associated with atypical **development** of widespread subcortical networks early in life....
- 🗨 Cite ➦ Share
- ☐ **FOXP genes, neural development, speech and language disorders.**
- 3 Takahashi H, Takahashi K, Liu FC.
Adv Exp Med Biol. 2009;665:117-29. doi: 10.1007/978-1-4419-1599-3_9.
PMID: 20429420 **Review.**
- Thus these genes appear to be involved in the **development** control of the central nervous system. Recently, **FOXP2**, a member of the Foxp subfamily, was identified as the first gene to be linked to an inherited form of **language** and speech disorder. The discovery of a **mutation** in **FOXP2** in a family with a speech and **language** disorder opened a new window to understanding the genetic cascades and neural

Breaking down the search query

- You are interested in studying **genetic variation** in **FOXP2** and its impact of **speech development**
- You would like to review all studies that have linked **mutations** in **FOXP2** to a **speech development disorder**
- You would also like to find the original study by Lai *et. al.* in 2001 that suggested that **FOXP2** is responsible for **speech development** in **humans**

FOXP2 Query Building

PubMed Advanced Search Builder

PubMed.gov
User Guide

Add terms to the query box

All Fields AND

Query box

(([ai[Author]] AND (foxp2[Title/Abstract])) AND ("2001"[Date - Publication] : "2001"[Date - Publication]))

History and Search Details

Search	Actions	Details	Query	Results	Time
#1	...	>	Search: (([ai[Author]] AND (foxp2[Title/Abstract])) AND ("2001"[Date - Publication] : "2001"[Date - Publication]))	1	05:38:19

Showing 1 to 1 of 1 entries

FOXP2 Query Building

> Nature. 2001 Oct 4;413(6855):519-23. doi: 10.1038/35097076.

A Forkhead-Domain Gene Is Mutated in a Severe Speech and Language Disorder

C S Lai ¹, S E Fisher, J A Hurst, F Vargha-Khadem, A P Monaco

Affiliations + expand

PMID: 11586359 DOI: 10.1038/35097076

Abstract

Individuals affected with developmental disorders of speech and language have substantial difficulty acquiring expressive and/or receptive language in the absence of any profound sensory or neurological impairment and despite adequate intelligence and opportunity. Although studies of twins consistently indicate that a significant genetic component is involved, most families segregating speech and language deficits show complex patterns of inheritance, and a gene that predisposes individuals to such disorders has not been identified. We have studied a unique three-generation pedigree, KE, in which a severe speech and language disorder is transmitted as an autosomal-dominant monogenic trait. Our previous work mapped the locus responsible, SPCH1, to a 5.6-cM interval of region 7q31 on chromosome 7 (ref. 5). We also identified an unrelated individual, CS, in whom speech and language impairment is associated with a chromosomal translocation involving the SPCH1 interval. Here we show that the gene FOXP2, which encodes a putative transcription factor containing a polyglutamine tract and a forkhead DNA-binding domain, is directly disrupted by the translocation breakpoint in CS. In addition, we identify a point mutation in affected members of the KE family that alters an invariant amino-acid residue in the forkhead domain. Our findings suggest that FOXP2 is involved in the developmental process that culminates in speech and language.

Save

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13 results

Search Results Page

RESULTS BY YEAR

TEXT AVAILABILITY

☐ Abstract
 ☐ Free full text
 ☐ Full text

ARTICLE ATTRIBUTE

☐ Associated data

ARTICLE TYPE

☐ Books and Documents
 ☐ Clinical Trial
 ☐ Meta-Analysis
 ☐ Randomized Controlled Trial

1

Equivalent missense variant in the **FOXP2** and **FOXP1** transcription factors causes distinct neurodevelopmental disorders.

Sollis E, Deriziotis P, Saitou H, Miyake N, Matsumoto N, Hoffer MJV, Ruivenkamp CAL, Alders M, Okamoto N, Bijlsma EK, Plomp AS, Fisher SE.

Hum Mutat. 2017 Nov;38(11):1542-1554. doi: 10.1002/humu.23303. Epub 2017 Aug 14.

PMID: 28741757

The closely related paralogues **FOXP2** and **FOXP1** encode transcription factors with shared functions in the **development** of many tissues, including the brain. ...We present here for the first time a direct comparison of the molecular and clinical consequences of the same **mutation** affecting the equivalent residue in **FOXP1** and **FOXP2**. ...

“ Cite

Share

2

Early neuroimaging markers of **FOXP2** intragenic deletion.

Liégeois FJ, Hildebrand MS, Bonthron A, Turner SJ, Scheffer IE, Bahlo M, Connelly A, Morgan AT.

Sci Rep. 2016 Oct 13;6:35192. doi: 10.1038/srep35192.

PMID: 27734906

Free PMC article.

FOXP2 is the major gene associated with severe, persistent, developmental speech and **language disorders**. While studies in the original family in which a **FOXP2 mutation** was found showed volume reduction and reduced activation in core **language** and speech networks, there have been no imaging studies of different **FOXP2** mutations. ...The hippocampus is implicated for the first time in **FOXP2** diseases. We conclude that **FOXP2** anomaly is either directly or indirectly associated with atypical **development** of widespread subcortical networks early in life ...

Free Full Text Articles

Review > Adv Exp Med Biol. 2009;665:117-29. doi: 10.1007/978-1-4419-1599-3_9.

FOXP Genes, Neural Development, Speech and Language Disorders

Hiroshi Takahashi ¹, Kaoru Takahashi, Fu-Chin Liu

Affiliations + expand

PMID: 20429420 DOI: 10.1007/978-1-4419-1599-3_9

FULL TEXT LINKS

ACTIONS

“ Cite

☆ Favorites

Other Options

Advanced

Create alert

Create RSS

User Guide

Save

Email

Send to

Sorted by: Best match

Display options

13 results

Your search was processed without automatic term mapping because

1

Case Reports

> Hum Mutat. 2017 Nov;38(11):1542-1554.

Epub 2017 Aug 14.

DISPLAY OPTIONS

Format

Summary

Abstract

Sort by

Best match

Per page

10

Other Options

The screenshot shows a web interface with a top navigation bar containing 'Save', 'Email', and 'Send to' buttons. Below this, there are two rows of buttons: the first row has 'Save', 'Email', and 'Send to' (highlighted with a blue border); the second row has 'Save', 'Email', and 'Send to'. A dropdown menu is open from the 'Email' button in the second row, listing 'Clipboard', 'My Bibliography', 'Collections', and 'Citation manager'. Below the dropdown, there are input fields for 'To:' (with 'com' visible) and 'Subject:'. At the bottom, there is a 'Format:' dropdown menu set to 'Summary' and two buttons: 'Send email' (blue) and 'Cancel'.

Sidebar...

- The advanced search pages that I have mentioned as part of PubMed are found on almost all the databases at the NCBI
- These databases can be searched by building up queries in a similar way as in PubMed

Summary

- Literature searching is less frustrating when:
 - You select a single, high quality resource such as PubMed to search
 - You break up your query into useful keywords
 - You make use of advanced search pages to build up a query
 - You look up MeSH terms that might make your search more specific